Sivakami Srinivasan

Data Analytics | Marketing Analytics | Operations Analytics Intern – Winter 2027

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PROFESSIONAL SUMMARY

Master of Data Analytics candidate (CGPA 3.67/4.33) seeking a Winter 2027 internship in Data Analytics, Marketing analytics, or Operations analytics. With 8+ years of firmware and software engineering experience (DDR5, WLAN, automotive HMI) in debugging, log analysis, and system optimization moving into data analytics with skills in statistical modeling (regression, classification, ANOVA, ARIMA/SARIMAX), mathematical optimization (PuLP, OR-Tools), Python, SQL, Tableau, and Power BI. Proven track record in executing hypothesis testing, forecasting, and turning statistical and optimization results intodashboards for decision-makers.

EDUCATION

Master of Data Analytics | University of Niagara Falls Canada **Expected 2027**
CGPA: 3.67 / 4.33

Bachelor of Engineering, Electronics & Communication | Rajalakshmi Engineering College, India **May 2012**
CGPA: 8.87 / 10.0

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-Learn, Statsmodels, XGBoost, LightGBM, PuLP, OR-Tools, Matplotlib), SQL (MySQL), R, Bash, C/C++

Data Visualization & BI: Tableau Desktop (Calculated Fields, Parameters, Geospatial Mapping, Dashboard Actions), Power BI (DAX, Data Modeling), Advanced Excel (PivotTables, Statistical Functions)

Statistical Modeling & Analytics: Regression Analysis (Linear/OLS, Logistic), Classification Models, Hypothesis Testing (ANOVA, Mann-Whitney U, Welch's t-test), Time Series Forecasting (ARIMA, SARIMAX), Statistical Significance Testing, K-Means Clustering

Optimization & Prescriptive Analytics: Integer Linear Programming (ILP), Linear Programming, Ant Colony Optimization (ACO), Decision Trees, Monte Carlo Simulation, Data Envelopment Analysis (DEA), Supply Chain & Operations Analytics

Engineering & Tools: Embedded C, C++, Core Java, Linux, OpenWrt, QNX RTOS, GDB, Git, Perforce, JIRA, Klocwork, Coverity, RISC-V, ARM

Data Engineering & Cloud: AWS (S3, Redshift, Glue, Athena, Lambda), ETL Pipelines, Relational Database Design (DDL, DML, DQL), Data Cleaning and Data Munging

ANALYTICS PROJECTS

Population Health Diabetes Classification & BMI OLS Regression | *Python, Statsmodels, Scikit-Learn*

- Developed an OLS regression pipeline on population survey data with ANOVA-based feature analysis, log-transformations for heteroscedasticity, and VIF checks for multicollinearity
- Built a classification pipeline on 456K+ CDC records using SMOTE oversampling and Random Forest, achieving ROC-AUC of 0.795 in predicting diabetes diagnoses

AI Industry Impact Analysis & PESTEL Strategic Framework | *Python, Statsmodels, Scikit-Learn*

- Built SARIMAX/ARIMA time series models to forecast monthly financial risk exposure and adoption timelines across 157,000+ patent records and cybersecurity incident logs
- Applied Random Forest classification to cyber-loss patterns and K-Means clustering to segment innovation fields and predict patent approval timelines

Vehicle Dealership Dashboard & Operational Analytics | Power BI, Advanced Excel

- Collaborated on an interactive sales dashboard using DAX and data modeling to show dealer inventory and performance trends.
- Structured market segment and performance views to support operational decision-making.

Additional Analytics Projects | Excel, MySQL, Python, R, Power BI

- **Mortgage Underwriting Risk Assessment & Capital Optimization (Python, MySQL, Tableau):** Built classification models (Logistic Regression, XGBoost, LightGBM) confirming DTI/CLTV as denial drivers; formulated an ILP model in PuLP to optimize portfolio return under a 3.5% denial-risk cap; designed executive Tableau dashboards.
- **Stochastic Supply Chain & Operations Routing Optimization (Python, OR-Tools, MySQL):** Extended operations research literature (Wiley, 2019) to incorporate stochastic weather variables for dynamic routing optimization.
- **Predictive Analytics for E-Waste Management (Python, Scikit-Learn, Pandas):** Co-developed Gradient Boosting regression models ($R^2 = 0.83$) predicting per-capita e-waste generation across 47 countries; contributed time-series forecasting and SHAP analysis.
- **Ontario School Board Achievement Analysis (Excel):** Led a team analyzing EQAO/OSSLT literacy outcomes against enrollment across 70+ boards; used non-parametric tests (Mann-Whitney U, Spearman's correlation) to identify a significant public vs. Catholic board performance gap.
- **Fire Incident Analytics (MySQL, Excel):** Cleaned and analyzed 17,000+ incident records with a team; log-transformed linear regression and Welch's t-tests showed a statistically significant impact of smoke spread on casualty rates.
- **Flight Delay Prediction (Python):** Co-developed Linear Regression and Random Forest models for arrival delays; compared accuracy using MAE, R-squared, and feature importance.
- **Customer Segmentation (R):** Applied linear regression and K-means clustering to segment customers and inform targeted marketing strategies.

CERTIFICATIONS

- AWS Certified Cloud Practitioner (in progress, expected Dec 2026)
- Google Data Analytics Professional Certificate (Google/Coursera, 2025)
- Google Analytics (2024); Python Mega Course (Udemy, 2023)

PROFESSIONAL EXPERIENCE

Firmware Engineer (Contract) | AMD Canada, Markham, ON **Sep 2024 – Nov 2025**

- Developed DDR5 PHY firmware on RISC-V, including initialization and calibration routines; executed numerical data processing and register calculations (CDD/ODT) to calibrate memory parameters
- Implemented training features compliant with JEDEC DDR5 standards
- Debugged issues in a pre-silicon design verification (DV) environment; supported team onboarding and code reviews

Senior Software Engineer (Contract, then Permanent) | Qualcomm, India **Jan 2019 – May 2023**

- Enhanced WLAN firmware (11ac/11ax) for Qualcomm chipsets on Q6 ARMv7 and OpenWrt; built Virtual AP state machine and host/hardware interaction features
- Led resolution of customer-critical bugs for enterprise clients (Cisco, Aruba, TP-Link) across connectivity, IoT, and security, parsing multi-gigabyte execution logs to find root causes
- Designed and executed unit tests and regression plans

Senior Software Engineer | Visteon, India **Feb 2017 – Jun 2017**

- Developed and maintained the HMI UI core stack for TFT instrument clusters; applied RGBA and CLUT models in image processing and collaborated on GUI animation and rendering
- Implemented diagnostic tools and resolved static analysis issues for MISRA C compliance

Software Engineer | Tata Elxsi, India **Jul 2012 – Feb 2015**

- Contributed to a DLNA framework (audio-video transport, content directory service) for automotive infotainment
- Built admin interfaces and notification-statistics modules in Java/C++ for an SDN research project
- Developed features for the HMI UI core stack for TFT instrument clusters while deputed at Visteon.